

Answer Key — Fair Testing and Recording

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **only one variable is changed; the rest are kept the same** — a fair test changes only one variable, keeping all else constant.
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- Q2** (c) **to check the result is not a one-off accident** — repeats that agree show the result is reliable.
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- Q3** (a) **so others can repeat the work and check it** — records let others reproduce and verify the result.
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- Q4** (a) **True** — surprising observations can be important clues.
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- Q5** (a) **Both A and R true, and R explains A** — new results naturally raise the next questions — science keeps evolving.
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- Q6** **systematic** — the careful step-by-step method is systematic investigation.
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- Q7** **reliable** — agreeing repeats make a result reliable.
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Short answer

- Q8** A test where only one variable is changed and all others kept the same, e.g. frying same-thickness puris in oil at different temperatures only.
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- Q9** (1) Lets you compare runs and spot patterns. (2) Lets others repeat the work and verify it.
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- Q10** Same dough thickness and size, same oil temperature (any two kept constant).
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Long / application

- Q11** **Change:** hole vs no hole. **Keep same:** thickness, size, flour, oil temperature, way of dropping. **Measure:** whether/how much it puffs (or height). **Reliable:** repeat several times for each case and compare.
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- Q12** Two variables changed (thickness *and* temperature), so the cause is unclear. Fix: keep oil temperature the same and change only thickness, repeating the trials.
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